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Publication and related bias in meta-analysis: Power of statistical tests and prevalence in the literature

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Abstract

Publication and selection biases in meta-analysis are more likely to affect small studies, which also tend to be of lower methodological quality. This may lead to "small-study effects," where the smaller studies in a meta-analysis show larger treatment effects. Small-study effects may also arise because of between-trial heterogeneity. Statistical tests for small-study effects have been proposed, but their validity has been questioned. A set of typical meta-analyses containing 5, 10, 20, and 30 trials was defined based on the characteristics of 78 published meta-analyses identified in a hand search of eight journals from 1993 to 1997. Simulations were performed to assess the power of a weighted regression method and a rank correlation test in the presence of no bias, moderate bias or severe bias. We based evidence of small-study effects on $P < 0.1$. The power to detect bias increased with increasing numbers of trials. The rank correlation test was less powerful than the regression method. For example, assuming a control group event rate of 20% and no treatment effect, moderate bias was detected with the regression test in 13.7%, 23.5%, 40.1% and 51.6% of meta-analyses with 5, 10, 20 and 30 trials. The corresponding figures for the correlation test were 8.5%, 14.7%, 20.4% and 26.0%, respectively. Severe bias was detected with the regression method in 23.5%, 56.1%, 88.3% and 95.9% of meta-anlyses with 5, 10, 20 and 30 trials, as compared to 11.9%, 31.1%, 45.3% and 65.4% with the correlation test. Similar results were obtained in simulations incorporating moderate treatment effects. However the regression method gave false-positive rates which were too high in some situations (large treatment effects, or few events per trial, or all trials of similar sizes). Using the regression method, evidence of small-study effects was present in 21 (26.9%) of the 78 published meta-analyses. Tests for small-study effects should routinely be performed in meta-analysis. Their power is however limited, particularly for moderate amounts of bias or meta-analyses based on a small number of small studies. When evidence of small-study effects is found, careful consideration should be given to possible explanations for these in the reporting of the meta-analysis. © 2000 Elsevier Science Inc. All rights reserved.

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1. Introduction

The importance of being systematic when reviewing the evidence that is available on the benefits and risks of medical intervention is widely recognized. Systematic reviews are prepared using reproducible strategies to avoid bias [1]. Results from individual studies are, when appropriate, combined in meta-analysis, a technique which has become increasingly popular in medical research [2]. Meta-analysis is not an infallible tool, however. Large multicenter trials have contradicted the results of earlier meta-analyses that were based on smaller trials [3–5]. Furthermore, meta-analyses addressing the same question have produced contradictory results [6,7].

The dissemination of findings from clinical trials is influenced by a host of factors which modify the probability that a study is included in a meta-analysis. If these factors are associated with trial results, then the meta-analysis will be biased. Publication bias, the tendency for increased publication rates among studies that show a statistically significant effect of treatment, has been documented repeatedly [8–10]. Published studies reporting statistically significant results are preferentially published in English [11], in journals indexed in Medline [8], are more likely to be cited by other authors [12,13] and are more likely to produce multiple publications [14,15]. Such "positive" studies are therefore more likely to be located for and included in meta-analyses, which may introduce bias.

Several aspects of trial quality have been shown to influence effect sizes. Inadequate concealment of treatment allocation resulting, for example, from the use of open random number tables, has been associated with larger treatment ef-

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fects [16,17]. Larger effects have also been found if investigators assessing outcomes knew what treatment patients had received [16,18], or if some participants, for example those not adhering to study medications, were excluded from the analysis [19]. Investigators sometimes undermine random allocation, for example by opening assignment envelopes or holding translucent envelopes up to a light bulb [20].

The smaller a study, the larger the treatment effect necessary for the results to be declared statistically significant. In addition, the greater investment of money and time in larger studies means that they are more likely to be of high methodological quality and published even if their results are negative. Bias in a systematic review may therefore be evident in an association between treatment effect and study size, and may be shown graphically in "funnel plots," scatter plots of the treatment effects estimated from individual studies on the horizontal axis against study size or standard error on the vertical axis which were first used in educational research and psychology [21]. The name "funnel plot" is based on the fact that the precision in the estimation of the underlying treatment effect will increase as the sample size of component studies increases. Effect estimates from small studies will therefore scatter more widely at the bottom of the graph, with the spread narrowing among larger studies. In the absence of bias the plot will resemble a symmetrical inverted funnel (see Fig. 1a).

Bias [e.g., because smaller studies showing no statistically significant effects (open circles in Fig. 1a) remain unpublished], will lead to an asymmetrical appearance of the funnel plot with a gap in the right bottom side of the graph (Fig. 1b). In this situation the combined effect from meta-analysis will overestimate the treatment's effect [5,22]. Such asymmetry might also result from the overestimation of treatment effects in smaller studies of lower methodological quality (Fig. 1c).

Funnel plot asymmetry cannot, however, be considered to be proof of bias in a meta-analysis. Heterogeneity between the treatment effects in different trials may lead to funnel plot asymmetry if the true treatment effect is larger in the smaller trials. For example, if a combined outcome is considered, substantial benefit may be seen only in patients at high risk for the component of the combined outcome which is affected by the intervention [23]. Trials conducted in high-risk patients will also tend to be smaller, because of the difficulty in recruiting such patients. Similarly, some interventions may have been implemented less thoroughly in larger trials, thus explaining the more positive results in smaller trials. This is particularly likely in trials of complex interventions in chronic diseases, such as rehabilitation after stroke or multifaceted interventions in diabetes mellitus. Different mechanisms which can lead to funnel plot asymmetry are summarized in Table 1.

We suggest the term "small-study effects" to describe a trend for the smaller studies in a meta-analysis to show larger treatment effects. While "small" trials are generally defined in relative terms (compared to the larger trials in a particular meta-analysis), a trial may be considered small in absolute terms if its sample size is too small to detect a clinically plausible effect. If small-study effects are present then there will be an association between treatment effect size and its standard error, so that the funnel plot will be asymmetrical (Fig. 1b and c), and further investigation of the reasons for these effects, including the potential for bias in the meta-analysis, is required.

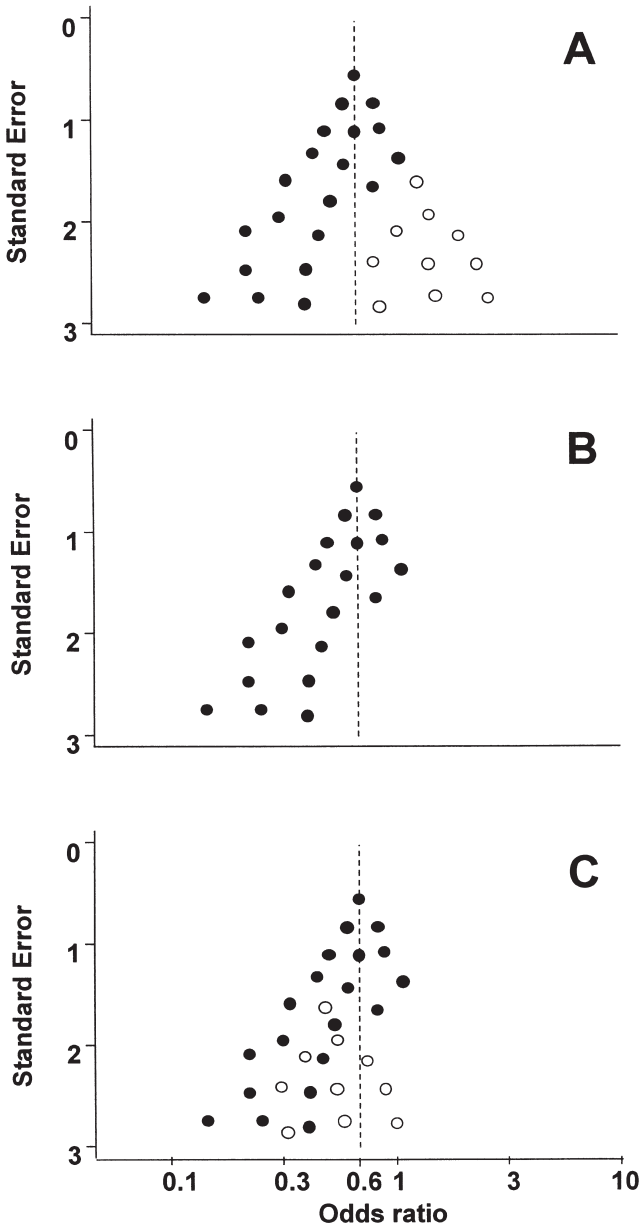


Fig. 1. Hypothetical funnel plots. (a) Symmetrical plot in the absence of bias (open circles indicate smaller studies showing no statistically significant effects); (b) asymmetrical plot in the presence of publication bias (smaller studies showing no statistically significant effects are missing); (c) asymmetrical plot in the presence of bias due to low methodological quality of smaller studies (open circles indicate small studies of inadequate quality whose results are biased toward larger effects).

The association between treatment effect size and its standard error is the basis of two statistical methods, a rank

Table 1
Potential sources of asymmetry in funnel plots

1. Reporting biases
Publication bias
Language bias
Citation bias
Multiple publication bias
2. True heterogeneity
Size of effect differs according to study size
Intensity of intervention
Differences in underlying risk
3. Data irregularities
Poor methodological design of small studies
Inadequate analysis
Fraud
4. Chance

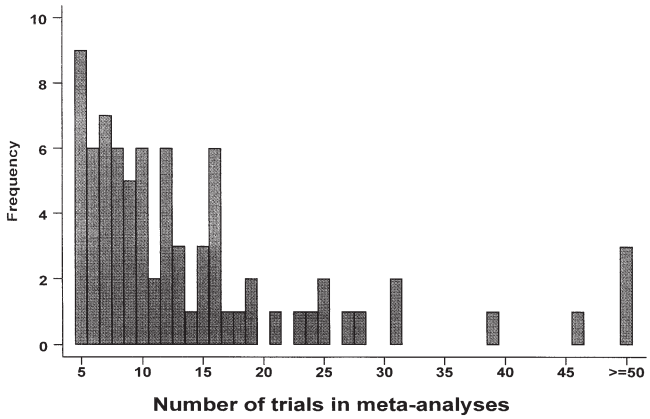


Fig. 2. Distribution of the number of trials in 78 meta-analyses identified in hand searches of leading medical journals.

correlation test [24] and a regression method [5], which have been proposed as a means of avoiding the subjectivity associated with visual assessment of funnel plots and which are described in more detail in the Methods section. However the validity of these methods has been questioned [25–27]. In the present study we examine the power of the two methods to detect small-study effects in circumstances typical of meta-analyses published recently, and identify situations in which the methods are not reliable. We then examine the evidence for small-study effects in meta-analyses published in leading journals.

2. Methods

2.1. Identification of published meta-analyses

To examine the characteristics of meta-analysis published in recent years, we searched four general medicine journals (Annals of Internal Medicine, BMJ, JAMA, Lancet) and four specialist journals (American Journal of Cardiology, Cancer, Circulation, Obstetrics and Gynecology), selected because a Medline search indicated that they publish many meta-analyses. Volumes from 1993 to 1997 were hand-searched for meta-analyses based on at least five trials with binary endpoints. Meta-analyses of community-randomized trials, which include atypically large numbers of individuals, were excluded. To obtain consistency across meta-analyses, endpoints were recoded if necessary, so that odds ratios below 1 indicated a beneficial effect of treatment. Of 95 eligible meta-analyses which were found, 78 (82%) reported the number of patients and events in each trial (1145 trials) and were included in the present study. Fig. 2 shows the distribution of the number of trials in each meta-analysis; 39 (50%) meta-analyses had 10 trials or fewer.

2.2. Definition of "typical" meta-analyses

We defined a set of hypothetical meta-analyses based on the characteristics of the 78 published meta-analyses. We grouped these according to the number of trials per meta-analysis: there were 9 (11.5%) meta-analyses with 5 trials, 42 (53.9%) with 6–14 trials, 16 (20.5%) with 15–24 trials, 6 (7.7%) with 25–35 trials and 5 (6.4%) meta-analyses with more than 35 trials. We defined four hypothetical meta-analyses containing $n = 5$, $n = 10$, $n = 20$, and $n = 30$ trials, respectively. To derive typical sizes (total number of patients) of trials within each group we ordered all the trials from all meta-analyses in each group by trial size. For the group with n trials, we chose trial sizes corresponding to the $1/(n + 1)$, $2/(n + 1)$, up to the $n/(n + 1)^{th}$ percentile, so that for the first group we chose five trial sizes corresponding to the 16.7^{th}, 33.3^{th}, 50th, 66.7^{th} and 83.3^{th} percentile, and so on. The trial sizes thus derived for the four typical meta-analyses are shown in Table 2.

2.3. Statistical tests for funnel plot asymmetry

Begg and Mazumdar [24] proposed an adjusted rank correlation test for the association between treatment effect size (t) and its variance v ($v = s^2$ where s is the standard error of t). The variances are stabilized by standardizing the effect sizes prior to performing the test. This is based on Kendall's tau, derived by enumerating the number of pairs of studies that are ranked in the same order with respect to

Table 2
Sample size of trials included in the four hypothetical meta-analyses which were used in simulations

Sample size	Meta-analysis based on 5 trials (No. of trials)	Meta-analysis based on 10 trials (No. of trials)	Meta-analysis based on 20 trials (No. of trials)	Meta-analysis based on 30 trials (No. of trials)
24–49	1	1	2	3
50–74	1	1	3	3
75–99	0	1	3	3
100–149	1	2	3	5
150–249	1	1	3	4
250–499	0	2	3	6
500–999	1	1	2	3
1,000–4,999	0	1	1	3

adjusted effect size and variance, Egger et al. [5] suggested a linear regression method in which the standard normal deviate (t/s) is regressed against precision ($1/s$). This latter method can be shown to be identical to a weighted regression of effect size t on standard error s ($t = b_0 + b_1 s$), where the weights are inversely proportional to the variance of the effect size. The greater the value of the bias coefficient b_1, the greater the evidence for small-study effects. Inverse variance weighting allows for between-study heterogeneity by including a multiplicative overdispersion parameter by which the treatment variance in each study is multiplied [28].

2.4. Simulations

Simulation analyses were used to assess the power (sensitivity) of the tests to detect bias, and to identify when they give false-positive results. The extent of bias was defined by assuming a linear relationship between treatment effect (log odds ratio) and its standard error (S.E.):

$$\log(\text{OR}) = \text{true treatment effect} + (\text{bias coefficient} \times \text{S.E. of log OR}).$$

The trials of magnesium treatment following myocardial infarction are a well-known example in which there is evidence of funnel-plot asymmetry and discrepancy between a meta-analysis and a subsequent large trial [29]. The estimated bias coefficient, using the regression method, for this set of trials was approximately -1. We defined this amount of bias as "severe," and did simulations with bias coefficients 0, -0.5, and -1.0 (defined as no, moderate and severe bias, respectively). To illustrate the amount of bias implied by these coefficients, assume a control group event rate of 20% and no treatment effect. For a trial with 40 participants per group, bias coefficients of -0.5 and -1.0 will lead to biased odds ratios of 0.75 and 0.53 respectively, while for a trial with 10,000 participants they will lead to odds ratios of 0.98 and 0.97; close to the true value of 1.

For simulations in which non-zero bias was assumed, we used an iterative procedure to calculate the actual treatment odds ratio, and hence the treatment group event rate, in each trial; based on the number of patients, control group event rate, treatment odds ratio in the absence of bias and bias coefficient. An iterative procedure is required because the odds ratio depends via the bias coefficient on the standard error, which in turn depends on the treatment odds ratio. For example, suppose that the true treatment effect (OR in the absence of bias) is 1, the control group event rate is 20% and that the trial has 40 participants per group. For bias coefficient -1 (severe bias) the equation

$$\log \text{OR} = \text{true treatment effect} + (\text{bias coefficient} \times \text{S.E. of log OR})$$

reduces to:

$$\log \text{OR} = \log(1) + (-1 \times \text{S.E. of log OR})$$

and so

$$\text{S.E. of log OR} = -\log \text{OR} \qquad (\text{since } \log(1) = 0).$$

The iterative procedure calculates the OR to be 0.53, so that $\log(\text{OR}) = \log(0.53) = -0.63$. To verify that this is correct, we calculate the expected number of individuals in each group. In the control group we expect a 20% disease rate so the odds are $0.2/0.8 = 0.25$ and the expected numbers are 8 with and 32 without disease. The odds in the treatment group are $0.25 \times 0.53 = 0.1325$. The disease rate in the treatment group is thus $0.1325/(1 + 0.1325) = 0.117$. The expected numbers are 4.68 with and 35.32 without disease. The standard error of the log OR is estimated by

$$\sqrt{(1/8) + (1/32) + (1/4.68) + (1/35.32)} = 0.63.$$

We based our choice of control group event rates on those found in published meta-analyses. The median control group event rate ranged from 8.9% to 18.1% in the 78 published meta-analyses. We used control group event rates of 5%, 10% and 20% in order to include this range. In simulations for which the expected number of events in a trial was very small (low control group event rate, large odds ratio, large bias coefficient), it was sometimes not possible to find a treatment group event rate for which the bias equation fitted exactly. We then used the event rate for which the equation fitted most closely.

Two sets of simulations were done. First, we examined the power of the methods to detect bias by simulating 10,000 meta-analyses for each "typical" meta-analysis, degree of bias and control group event rate (1,080,000 simulations). These simulations assumed odds ratios of 1 (no true treatment effect), 0.5 and 0.25 (protective treatment effects). In each simulation the number of events in each group in each trial was generated randomly using the binomial distribution. We calculated the power (the proportion of simulations which gave evidence for bias) for each method, in each set of simulations.

It has been claimed that the regression method may give false-positive results in the absence of bias [26]. We examined this in further simulations, all done with bias coefficient 0, first based on the "typical" meta-analyses with an extreme treatment odds ratio of 0.1 (12,000 simulations), and second based on the characteristics of the 78 published meta-analyses. For each of these meta-analyses we derived the overall treatment effect (from fixed-effects meta-analysis), the treatment and control group sizes in each trial, and observed control group event rate in each trial. These were then used to simulate 10,000 sets of results per published meta-analysis, assuming no bias (780,000 simulations).

Finally, we examined the prevalence of small-study effects using the reported data from each of the 78 published meta-analyses. All analyses were done using the general statistical package Stata (Stata Corporation, College Station, TX, USA [30]. In all analyses, trials with no events were omitted; otherwise if any cell in the standard two-by-

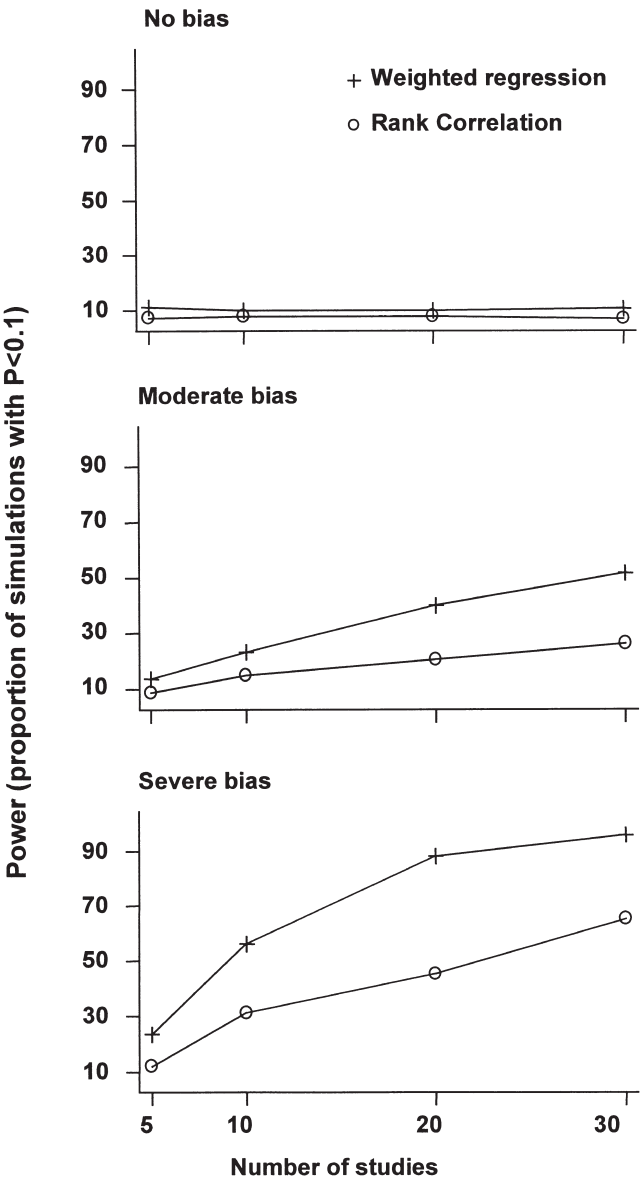


Fig. 3. Power of two methods to detect small-study effects ($P < 0.1$), for different amounts of underlying bias. Simulations were based on four hypothetical meta-analyses which are typical of those published in leading journals from 1993 to 1997.

two table contained zero, we added 0.5 to all four cells. We based evidence for bias on $P < 0.1$, as used in previous analyses [5,31].

3. Results

3.1. Simulations based on hypothetical, "typical" meta-analyses

Fig. 3 shows the power of the tests to detect bias ($P < 0.1$) derived from simulations with a control group event rate of 20% and assuming no treatment effect. Table 3 shows the results from the same analyses for control group event rates of 5%, 10% and 20%. In the absence of bias, the proportion of false-positive results was close to the correct value of 10%. The power increased with increasing numbers of trials but was low for meta-analyses with less than 10 trials, and for moderate bias (bias coefficient −0.5). The correlation test was less sensitive than the regression method, particularly when the event rate was low.

Table 3
Power (percent of simulations with $P < 0.1$) of two tests for small-study effects when the true treatment OR = 1, for different amounts of simulated bias and different control group event rates

	Power (%)			
Bias	5 Trials	10 Trials	20 Trials	30 Trials
None				
Control group event rate 5%				
Weighted regression	9.8	10.5	10.7	10.1
Rank correlation	6.2	7.7	7.0	6.2
Control group event rate 10%				
Weighted regression	10.9	10.3	10.8	10.7
Rank correlation	7.2	8.5	7.5	7.0
Control group event rate 20%				
Weighted regression	11.3	10.1	10.0	10.6
Rank correlation	7.3	7.7	7.7	6.9
Moderate				
Control group event rate 5%				
Weighted regression	11.9	21.8	40.3	48.7
Rank correlation	5.5	11.0	14.5	19.4
Control group event rate 10%				
Weighted regression	13.8	23.6	38.5	49.7
Rank correlation	7.4	14.3	17.5	24.5
Control group event rate 20%				
Weighted regression	13.7	23.5	40.1	51.6
Rank correlation	8.5	14.7	20.4	26.0
Severe				
Control group event rate 5%				
Weighted regression	17.7	58.6	91.3	96.9
Rank correlation	4.5	15.1	13.5	28.0
Control group event rate 10%				
Weighted regression	22.8	57.5	90.1	96.4
Rank correlation	8.9	27.3	36.5	57.0
Control group event rate 20%				
Weighted regression	23.5	56.1	88.3	95.9
Rank correlation	11.9	31.1	45.3	65.4

Table 4 and Table 5 show the power of the tests derived from simulations which included true treatment odds ratios of 0.5 and 0.25, respectively. In the absence of bias, the false-positive rate for the rank-correlation method was generally too low, particularly for treatment OR 0.25. The power of the regression method was generally similar to its power when there was no treatment effect (OR = 1, Table 3). In most simulations including a treatment effect the rank correlation method had considerably lower power than the regression method.

Table 6 shows false-positive rates from simulations with bias coefficient 0 but with an extreme treatment OR of 0.1. For control group event rate 5% both methods gave false-positive rates greatly in excess of the correct value of 10%. For control event rates of 10% and 20% the false-positive rates for the regression method were close to the correct value. The rank cor-

Table 4
Power (percent of simulations with $P < 0.1$) of two tests for small-study effects when the true treatment OR = 0.5, for different amounts of stimulated bias and different control group event rates

	Power (%)			
Bias	5 Trials	10 Trials	20 Trials	30 Trials
None				
Control group event rate 5%				
Weighted regression	12.7	10.0	10.5	9.8
Rank correlation	5.8	5.8	5.3	5.8
Control group event rate 10%				
Weighted regression	10.7	10.4	10.0	10.1
Rank correlation	5.9	7.1	6.5	7.2
Control group event rate 20%				
Weighted regression	10.5	10.1	10.8	10.7
Rank correlation	6.6	8.2	8.8	8.6
Moderate				
Control group event rate 5%				
Weighted regression	16.7	25.8	41.8	53.8
Rank correlation	3.7	4.4	2.8	4.5
Control group event rate 10%				
Weighted regression	14.3	23.9	41.5	55.0
Rank correlation	4.9	9.9	14.6	20.8
Control group event rate 20%				
Weighted regression	15.8	26.9	46.7	58.5
Rank correlation	7.8	15.7	26.9	36.1
Severe				
Control group event rate 5%				
Weighted regression	20.0	58.4	87.5	96.6
Rank correlation	6.3	2.7	10.8	3.9
Control group event rate 10%				
Weighted regression	25.4	68.0	93.7	98.5
Rank correlation	3.9	16.1	13.7	30.6
Control group event rate 20%				
Weighted regression	27.3	64.7	93.4	98.3
Rank correlation	9.9	32.9	50.9	71.8

Table 5
Power (percent of simulations with $P < 0.1$) of two tests for small-study effects when the true treatment OR = 0.25, for different amounts of simulated bias and different control group event rates

	Power (%)			
Bias	5 Trials	10 Trials	20 Trials	30 Trials
None				
Control group event rate 5%				
Weighted regression	10.2	8.4	10.4	9.1
Rank correlation	4.7	3.5	2.1	1.9
Control group event rate 10%				
Weighted regression	9.3	8.7	8.5	9.2
Rank correlation	4.2	3.9	3.3	3.6
Control group event rate 20%				
Weighted regression	10.2	11.0	10.6	10.9
Rank correlation	5.9	6.8	8.0	9.4
Moderate				
Control group event rate 5%				
Weighted regression	15.4	16.5	25.6	33.6
Rank correlation	13.5	6.7	26.5	20.9
Control group event rate 10%				
Weighted regression	11.7	24.2	38.6	51.5
Rank correlation	3.5	3.3	2.2	2.6
Control group event rate 20%				
Weighted regression	14.1	28.0	45.7	61.0
Rank correlation	4.8	12.4	20.6	31.6
Severe				
Control group event rate 5%				
Weighted regression	22.9	23.4	41.2	66.4
Rank correlation	35.1	28.2	82.7	89.3
Control group event rate 10%				
Weighted regression	12.7	57.7	84.0	96.6
Rank correlation	6.3	2.6	13.9	5.4
Control group event rate 20%				
Weighted regression	24.7	70.6	95.6	99.1
Rank correlation	4.4	20.7	22.9	45.1

relation method gave false-positive rates that were much too low when the control group event rate was 20%.

3.2. Simulations based on the characteristics of the 78 published meta-analyses

No bias was present in these simulations and the false-positive rate should therefore be close to 10%. Table 7 shows that the proportion of simulations in which the false-positive rate was within 2% of the correct value was 74.4% for the regression method but only 37.2% for the rank correlation method. This was because the rank correlation method often had false-positive rates which were too low. The number of simulations giving false-positive rates of 14% or more was 8 (10.3%) for the regression method and 5 (6.4%) for the rank correlation method. Detailed examination of these results showed that anomalous results occurred when there was a substantial treatment effect, or the average number of events per trial in one or both groups was small, or all trials in the meta-analysis had similar sample sizes. For example, the three highest false-positive rates for the regression method were 23.1%, 21.2% and 20.6%. In these simulations the odds ratios were 0.32, 0.14 and 0.16 respectively, and the mean expected number of events per trial in the treatment group were 3.5, 1.0 and 7.3 respectively.

3.3. Evidence for small-study effects in published meta-analyses

Using the regression method the number of meta-analyses with evidence for small-study effects was 21 (26.9%). The corresponding number for the rank correlation method

Table 6
False-positive rates (percent of simulations with $P < 0.1$), in the absence of bias, for an extreme true treatment OR of 0.1 and different control group event rates

	False-positive rate (%)			
	5 Trials	10 Trials	20 Trials	30 Trials
Control group event rate 5%				
Weighted regression	14.4	14.8	33.5	27.1
Rank correlation	14.2	24.3	38.4	48.2
Control group event rate 10%				
Weighted regression	6.6	6.6	12.0	8.8
Rank correlation	7.8	6.3	8.6	6.9
Control group event rate 20%				
Weighted regression	7.1	7.4	7.0	6.9
Rank correlation	3.7	2.7	1.6	1.3

Table 7
False-positive rates (percent of simulations with $P < 0.1$) in the absence of bias, from simulations based on the characteristics of 78 published meta-analyses

False-positive rate (% of simulations with $P < 0.1$)	Weighted regression method *n* (%)	Rank correlation method *n* (%)
<6	0	13 (16.7)
6–7.99	2 (2.6)	23 (29.5)
8–11.99	58 (74.4)	29 (37.2)
12–13.99	10 (12.8)	8 (10.3)
⩾14	8 (10.3)	5 (6.4)

was 10 (12.8%), only marginally more than the expected false-positive rate. Funnel plots for the 21 meta-analyses identified using the regression method are shown in Fig. 4 and the references are given in the Appendix. Of these, the false-positive rate (using the regression method) in simulations replicating the meta-analysis was above 14% in 2 (numbers 2 and 14 in Fig. 4). The effect of treatment appeared more beneficial in smaller studies (bias coefficient <0) for 54 (69.2%) of the 78 published meta-analyses. The average magnitude of the bias coefficient (combined using fixed-effects meta-analysis) for the 78 meta-analyses was −0.54 (95% CI −0.65–−0.42). For such a bias coefficient, assuming a control group event rate of 20% and no treatment effect, a trial with 40 participants per group will give a biased odds ratios of 0.71 instead of the correct value of 1.

4. Discussion

The fallibility of meta-analysis is not surprising given the biases which may be introduced in the process of locating and selecting studies [8,11,12,32], and the often inadequate quality of component studies [16,17]. These biases are more likely to affect small studies, which also tend to be of lower methodological quality. Several biases, as well as other sources of heterogeneity, may act simultaneously in a given meta-analysis and their relative contribution cannot always be disentangled.

It has long been recognized that biases which result in an association between treatment effects and sample size are potentially diagnosable, but the use of statistical tests to detect such "small-study effects" has not been evaluated. We compared two methods which examine the association between the observed effect and its standard error among the studies in a meta-analysis. Standard error is a measure of study size because for larger studies the effect sizes will have smaller standard errors, and is preferable to sample size in comparing studies because the power of a study depends on the number of events; a study with 100,000 patients and 10 events is less powerful than a study with 1000 patients and 100 events.

The number of trials, their sizes, and the event rates in intervention and control groups will influence the performance of the tests [24,27]. To evaluate their performance in "the real world" we based our simulations on meta-analyses published recently in leading medical journals. We found that the regression method was more powerful than the rank correlation test in most situations. The power of both methods was low for meta-analyses based on 10 or fewer trials, or in the absence of severe bias. As expected, the power of the methods increased with increasing numbers of trials and increasing event rates. We found that in the absence of bias the rank correlation method produced false-positive rates which were incorrectly low. This is in accordance with the simulation analyses reported by Begg and Mazumdar [24].

We identified circumstances in which the tests may give false-positive results. The principal reason is that when the treatment effect is large the distribution of log odds ratio or log relative risk is asymmetric. An example is shown in Fig. 5, for 100 individuals per group, a control group event rate of 20% and treatment group event rate 5% (OR = 0.21). Here, the sampling distribution of the log odds ratio is left skewed, which means that, by chance alone, log odds ratios less than log (0.21) are more likely to occur than log odds ratios greater than log (0.21). The smaller log odds ratios will also have larger standard errors (because they are based on fewer events). This will lead to funnel plot asymmetry and, potentially, to spurious associations between treatment effect size (log odds ratio) and standard error in the corresponding statistical tests.

We found spurious associations when there was a substantial treatment effect, or all trials had low event rates, or the studies were all of similar sizes. Our simulations based on published meta-analyses indicate that for the regression method, this problem arises in some 10% of meta-analyses; mainly when there is a substantial treatment effect and the average number of events per trial is very low. In particular, the method is not suitable for meta-analyses of rare outcomes such as adverse events. Taken together, these results show that the regression method is appropriate and reasonably powerful in the situations where meta-analysis generally makes sense; in estimating moderate treatment effects, based on a reasonable number of trials. However it should only be used if there is clear variation in trial sizes, with one or more trials of medium or large size.

Our approach to defining bias differs from that used in a number of studies in which the effect of publication bias is modeled directly via "selection models" which assume that the probability that a study is published (and thus included in a meta-analysis) depends on the P-value for the estimated treatment effect [24,33,34]. In contrast, we have assumed that bias leads to linear association between log odds ratio and its standard error, and evaluated the tests on the basis of this assumption. Small-study effects may arise not only because of publication bias but because of other factors, for example if smaller studies are of lower methodological quality.

Our simulations assumed that small-study effects were manifested in a linear association between treatment effect size (log odds ratio) and its standard error. The rank correla-

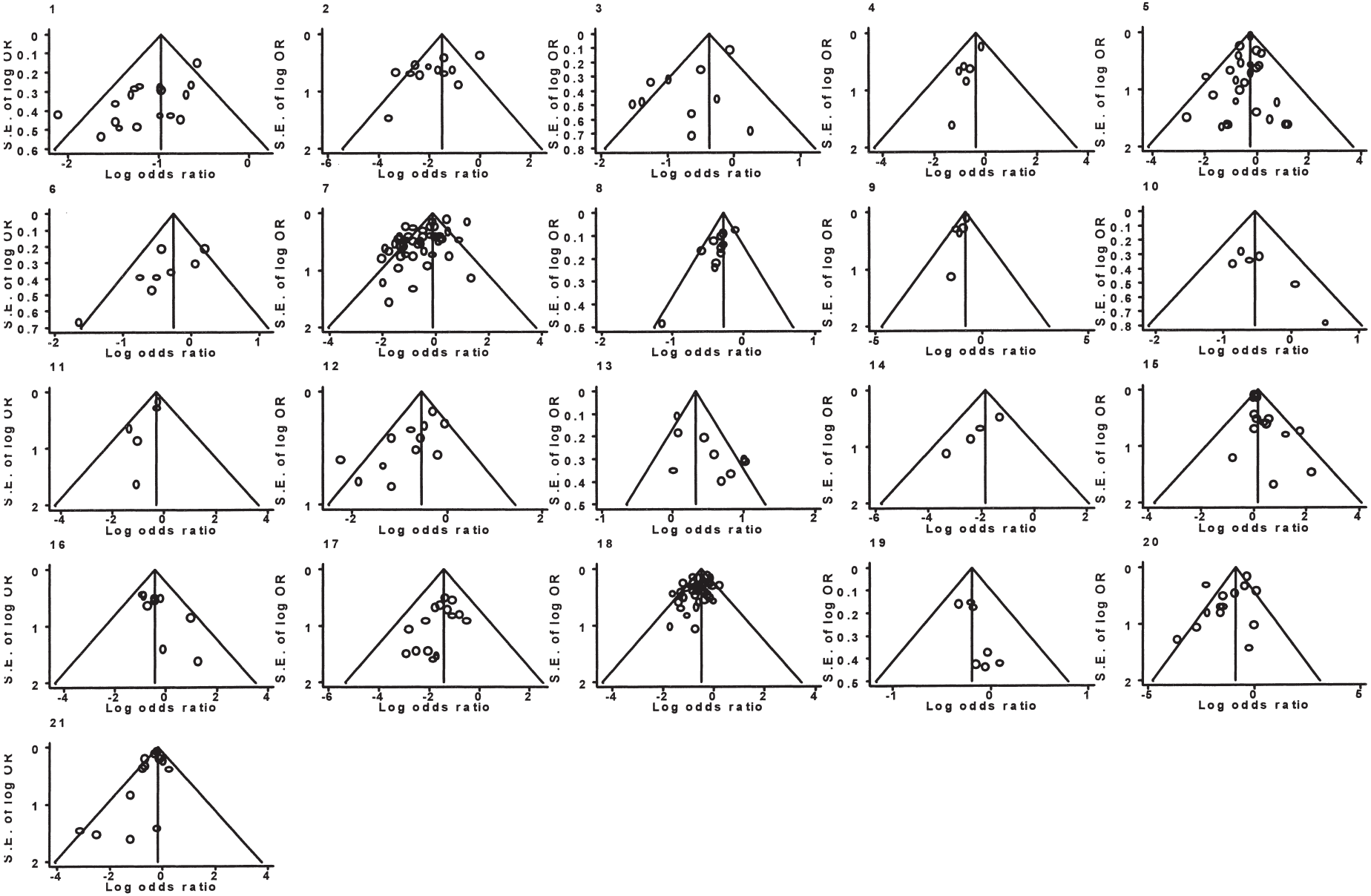


Fig. 4. Funnel plots for 21 meta-analyses with evidence ($P < 0.1$) of small-study effects from the weighted regression method. Graph numbers (top left of each funnel plot) refer to trial number in the Appendix.

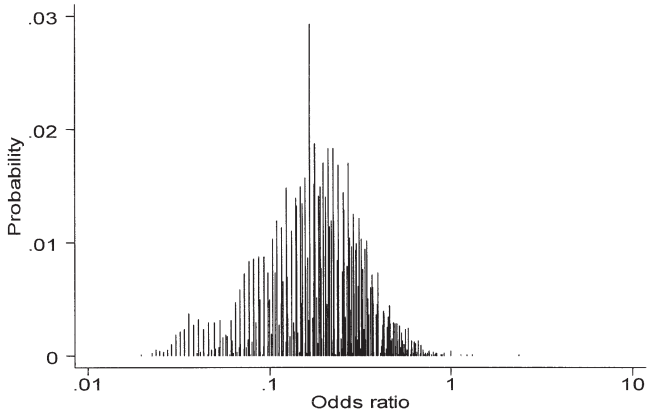


Fig. 5. Sampling distribution of the odds ratio (log scale) for a trial with 100 individuals per group, control group event rate 20% and treatment group event rate 5% (OR = 0.21).

tion test might have been expected to perform better in comparison to the regression method in situations where the true association is not in fact linear. However our results indicate that the regression method performs better in practice; the rank correlation test found evidence for small-study effects in only 10 (12.8%) of the 78 meta-analyses compared to 21 (26.9%) found by the regression method.

In this study, 50% of published meta-analyses were based on 10 trials or fewer and we cannot, therefore, exclude with confidence the presence of bias in meta-analyses that tested negative. Indeed, smaller studies tended to show greater treatment effects than larger studies in 69% of the meta-analyses. Despite the limited power of the regression method, we found evidence of small-study effects in 21 (27%) of the published meta-analyses. However it would not be reasonable to infer that all these were affected by bias. We would expect about 8 (10% of 78) to have P-values <0.1 even in the absence of any small-study effects. Further, when attempting to explain small-study effects the possibility of bias should be considered along with the possibility that genuine heterogeneity in treatment effects was associated with study size (Table 1). Whatever the source of the small-study effects, their presence raises the question of whether it is appropriate to report a single summary treatment effect as the result of the meta-analysis.

Several studies have compared the results from large trials with those from meta-analyses of smaller trials and assessed the degree of agreement between the two groups of trials [35]. Findings from such studies are influenced by the definition of "large" trials and measures of agreement used, but it is noteworthy that the rates of disagreement were comparable to our estimates of the prevalence of small-study effects. One study found that funnel plot asymmetry was frequently present when there was disagreement between small and large trials [22].

In addition to providing a test for small-study effects, the regression coefficient represents a numerical estimate of the direction and amount of bias present in a given situation, and allows comparisons between groups of meta-analyses in "meta-epidemiological" research [25]. We used this approach to show that small-study effects may be less prevalent among reviews published in the Cochrane Database of Systematic Reviews than for meta-analyses published in high-impact journals [5]. The greater methodological rigor, in particular the more comprehensive literature searches, of Cochrane reviews [36], may mean that publication bias is reduced so that funnel plots are more symmetric.

Explanations for heterogeneity between studies may be examined using "meta-regression" models [37]. The regression method evaluated here is a meta-regression of treatment effect on standard error using a multiplicative overdispersion parameter (in the terminology of Thompson and Sharp [28]) to allow for heterogeneity between studies. Meta-regression models usually assume an additive between-study component of variance [28]. When there is underdispersion (less between-trial heterogeneity than expected at random) then the overdispersion parameter will be estimated to be less than 1 and this will make the regression method more sensitive to associations between effect size and standard error than standard meta-regression models. Funnel plots 4 and 19 in Fig. 4 are examples of such underdispersion.

An extension of the methods described here would be to use meta-regression models to investigate alternative explanations for between-study heterogeneity, by including the standard error of the treatment estimate with other study characteristics. For example, the simultaneous effects of study size and components of methodological quality could be examined [38]. More complex models are needed to examine whether underlying risk is associated with treatment effect [39,40]. These could also be extended to investigate underlying risk and standard error as competing explanations for between-study heterogeneity.

In conclusion, the possibility that the results of a meta-analysis may be influenced by small-study effects should routinely be examined, both graphically and statistically. Statistical tests for funnel plot asymmetry are required because visual inspection is subjective, and necessary to quantify the evidence for asymmetry. The regression method for detection of funnel plot asymmetry performs better than the rank correlation test. However its power is limited unless substantial bias is present and the number of trials is 10 or more. Further, it should only be used if there is clear variation in trial sizes, with one or more trials of medium or large size. Finally, the presence of small-study effects is not proof of bias, but should alert meta-analysts to the need for further investigation of their causes.

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Appendix

Published meta-analyses with evidence ($P < 0.1$) for publication or related bias.

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